

Dry Run Exercises : Solutions

Unit 5 Arrays

Complete the dry run table for each fragment of code given in the questions below. **Give** the value(s) printed out when each is executed.

Follow the module dry run style guide in your answers.

1.

```
int [] numbers = {4,5,2,5,7};           // L1

for(int i = 1; i<=2; i=i+1)             // L2
{
    System.out.println(numbers[i]);     // L3
}
```

Complete the dry run table (you may not need all rows):

LINE	numbers					i	i <= 2 ?	printed
	[0]	[1]	[2]	[3]	[4]			
L1	4	5	2	5	7			
L2						1	1<=2 ? true	
L3								5
L2						2	2<=2 ? true	
L3								2
L2						3	3<=2 ? false	

Write EXACTLY what is printed here

5
2

2.

```
int [] numbers = {3,4,5,6}; // L1

for(int i = 0; i<=2; i=i+1) // L2
{
    System.out.println(numbers[i]); // L3
}
System.out.println(0); // L4
```

Complete the dry run table (you may not need all rows):

LINE	numbers				i	i <= 2 ?	printed
	[0]	[1]	[2]	[3]			
L1	3	4	5	6			
L2					0	0<=2 ? true	
L3							3
L2					1	1<=2 ? true	
L3							4
L2					2	2<=2 ? true	
L3							5
L2						3<=2 ? false	
L4							0

Write EXACTLY what is printed here

3
4
5
0

```

3.
int [] numbers = {3,4,5,6,7};           // L1

for(int i = 0; i<4; i=i+1)              // L2
{
    System.out.println(numbers[i]);      // L3
}

```

Complete the dry run table (you may not need all rows):

LINE	numbers					i	i < 4 ?	printed
	[0]	[1]	[2]	[3]	[4]			
L1	3	4	5	6	7			
L2						0	0<4 ? true	
L3								3
L2						1	1<4 ? true	
L3								4
L2						2	2<4 ? true	
L3								5
L2						3	3<4 ? true	
L3								6
L2						4	4<4 ? false	

Write EXACTLY what is printed here

3
4
5
6

```

4.
int [] numbers = {3,4,5,6,7};           // L1

for(int i = 4; i>0; i=i-1)              // L2
{
    System.out.println(numbers[i]);      // L3
}

```

Complete the dry run table (you may not need all rows):

LINE	numbers					i	i > 0 ?	printed
	[0]	[1]	[2]	[3]	[4]			
L1	3	4	5	6	7			
L2						4	4>0 ? true	
L3								7
L2						3	3>0 ? true	
L3								6
L2						2	2>0 ? true	
L3								5
L2						1	1>0 ? true	
L3								4
L2						0	0>0 ? false	

Write EXACTLY what is printed here

7
6
5
4

```

5.
int [] numbers = {4,5,2,5,7};           // L1

for(int i = 1; i<=3; i=i+1)             // L2
{
    if (numbers[i] == 5)                 // L3
    {
        System.out.println(i);          // L4
    }
}

```

Complete the dry run table (you may not need all rows):

LINE	numbers					i	i <= 3 ?	numbers[i] == 5 ?	printed
	[0]	[1]	[2]	[3]	[4]				
L1	4	5	2	5	7				
L2						1	1<=3 ? true		
L3								5 == 5 ? true	
L4									1
L2						2	2<=3 ? true		
L3								2 == 5 ? false	
L2						3	3<=3 ? true		
L3								5 == 5 ? true	
L4									3
L2						4	4<=3 ? false		

Write EXACTLY what is printed here

1
3

6.

```
int [] numbers = {4,5,2,5,7};           // L1

for(int i = 1; i<=2; i=i+1)             // L2
{
    numbers[i] = 6;                     // L3
}
```

Complete the dry run table (you may not need all rows):

LINE	numbers					i	i <= 2 ?
	[0]	[1]	[2]	[3]	[4]		
L1	4	5	2	5	7		
L2						1	1<=2 ? true
L3		6					
L2						2	2<=2 ? true
L3			6				
L2						3	3<=2 ? false

Write the final value of the array here in the form eg {4,5,2,5,7}

{4,6,6,5,7}

```

7.
int [] a = {3,4,5,6};           // L1

for(int i = 0; i<=2; i=i+1)     // L2
{
    a[i] = i;                   // L3
}

```

Complete the dry run table (you may not need all rows):

LINE	a				i	i <= 2 ?
	[0]	[1]	[2]	[3]		
L1	3	4	5	6		
L2					0	0<=2 ? true
L3	0					
L2					1	1<=2 ? true
L3		1				
L2					2	2<=2 ? true
L3			2			
L2						3<=2 ? false

Write the final value of the array here in the form eg{4,5,2,5,7}

{0,1,2,6}

8.

```
int [] a = {3,4,5,6,7};           // L1

for(int i = 0; i<4; i=i+1)        // L2
{
    a[i] = a[i+1];                // L3
}
```

Complete the dry run table (you may not need all rows):

LINE	a					i	i < 4 ?
	[0]	[1]	[2]	[3]	[4]		
L1	3	4	5	6	7		
L2						0	0<4 ? true
L3	4						
L2						1	1<4 ? true
L3		5					
L2						2	2<4 ? true
L3			6				
L2						3	3<4 ? true
L3				7			
L2						4	4<4 ? false

Write the final value of the array here in the form eg {4,5,2,5,7}

{4,5,6,7,7}

9.

```
int [] b = {3,4,5,6,7};           // L1

for(int i = 4; i>0; i=i-1)        // L2
{
    b[i] = b[i-1];                // L3
}
```

Complete the dry run table (you may not need all rows):

LINE	b					i	i > 0 ?
	[0]	[1]	[2]	[3]	[4]		
L1	3	4	5	6	7		
L2						4	4>0 ? true
L3					6		
L2						3	3>0 ? true
L3				5			
L2						2	2>0 ? true
L3			4				
L2						1	1>0 ? true
L3		3					
L2						0	0>0 ? false

Write the final value of the array here in the form eg {4,5,2,5,7}

{3,3,4,5,6}

10.

```
int [] numbers = {4,5,2,5,7};           // L1

for(int i = 1; i<=3; i=i+1)             // L2
{
    if (numbers[i] == 5)                 // L3
    {
        numbers[i] = numbers[i] + 1;    // L4
    }
}
```

Complete the dry run table (you may not need all rows):

LINE	numbers					i	i <= 3 ?	numbers[i] == 5 ?
	[0]	[1]	[2]	[3]	[4]			
L1	4	5	2	5	7			
L2						1	1<=3 ? true	
L3								5 == 5 ? true
L4		6						
L2						2	2<=3 ? true	
L3								2 == 5 ? false
L2						3	3<=3 ? true	
L3								5 == 5 ? true
L4				6				
L2						4	4<=3 ? false	

Write the final value of the array here in the form eg {4,5,2,5,7}

{4,6,2,6,7}